Automatic License Plate Recognition using Contour Detection and K-Nearest Neighbours

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Abstract

With increasingly large numbers of cars making their way onto the road today, modern cities must establish automated systems for traffic management and scheduling. In this paper, we propose a system that allows automatic localization and recognition of vehicle license plates using a number of image processing techniques and machine learning algorithms. An effort is also made to extend the system to work with a real-time video stream in order to monitor traffic in movement. We achieve encouraging results and believe that such a system, if put into practice, can greatly reduce the burden on the people manning Lahore city's traffic management systems.

1 Introduction

Automatic License Plate Recognition (ALPR) is a surveillance method that uses Optical Character Recognition (OCR) on images of vehicles to read their registration plates. It plays an important role in numerous applications such as unattended parking lots, security control of restricted areas, traffic law enforcement, congestion pricing, and automatic toll collection. Additionally, the data collected by such a system over long period of time may help local authorities gain insight into traffic flow patterns in a particular area by looking at car park usage, pedestrian crossing usage, number of vehicles along a road, areas of low and high congestion, frequency, and location and cause of road works.

Most developed countries already have in place complex systems for license plate recognition. In the United States, Mobile ANPR is widely used by law enforcement agencies at the city, county, state and federal level. A 2012 report by the Police Executive Research Forum revealed that 71% of all US police departments employ ANPR in one form or another. Such systems are fast becoming a significant component of municipal predictive policing strategies and intelligence gathering, as well as for recovery of stolen vehicles, identification of wanted felons, and revenue collection from individuals who are delinquent on city or state taxes or fines. Successfully recognized plates may be matched against crime databases. In addition to real-time surveillance, the data collected (images, dates, times and GPS coordinates) can be stockpiled in order to place a suspect at a scene, assist in witness identification, pattern recognition or the tracking of individuals.

Due to different working environments, ANPR techniques vary from application to application. Of the various working conditions, outdoor scenes and non-stationary backgrounds may be the two factors that most influence the quality of scene images acquired and in turn the complexity of the techniques needed. In an outdoor environment, illumination not only changes slowly as daytime progresses, but may change rapidly due to changing weather conditions and passing objects. A dynamic scene image may contain multiple license plates or no license plate at all. Moreover, when they do appear in an image, license plates may have arbitrary sizes, orientations and positions. And, if complex backgrounds are involved, detecting license plates can become quite a challenge.

2 Related Work

There are a number of primary algorithms that most systems use for identifying a license plate:

1. Plate localization – responsible for finding and isolating the plate on the picture.
2. Plate orientation and sizing – compensates for the skew of the plate and adjusts the dimensions to the required size.
3. Normalization – adjusts the brightness and contrast of the image.
4. Character segmentation – finds the individual characters on the plates.
5. Optical character recognition – classifies the detected characters into one of 36 alphanumeric classes.

Typically, an ANPR process consists of two main stages: 1) locating license plates and 2) identifying license numbers.

In the first stage, license plate candidates are determined based on the features of license plates. Features commonly employed have been derived from the license plate format and the alphanumeric characters constituting license numbers [1]. The features regarding license plate format include shape, symmetry [3], eight-to-width ratio [7], colour [8], texture [2], spatial frequency [9], and variance of intensity values [10]. Character features include gradient magnitudes [4], aspect ratio of characters [6], distribution of intervals between characters [5], and alignment of characters [11], [12].

The license plate candidates determined in the localization stage are examined in the license number recognition stage. There are two major tasks involved in the recognition stage: character separation and character recognition. Character separation may be accomplished by such techniques as projection [13], morphology [19], relaxation labelling [16], connected components [14], and blob colouring [17]. For character recognition, one might employ genetic algorithms [18], support vector machine [19], artificial neural networks [20], or Markov processes [15].

3 Methodology

Our methodology employs a variety of different image processing techniques and machine learning algorithms in order to accurately localize license plates from images of vehicles and correctly classify individual characters on those plates to give the registration number.

Figure 1 on the right shows the operation of our system.

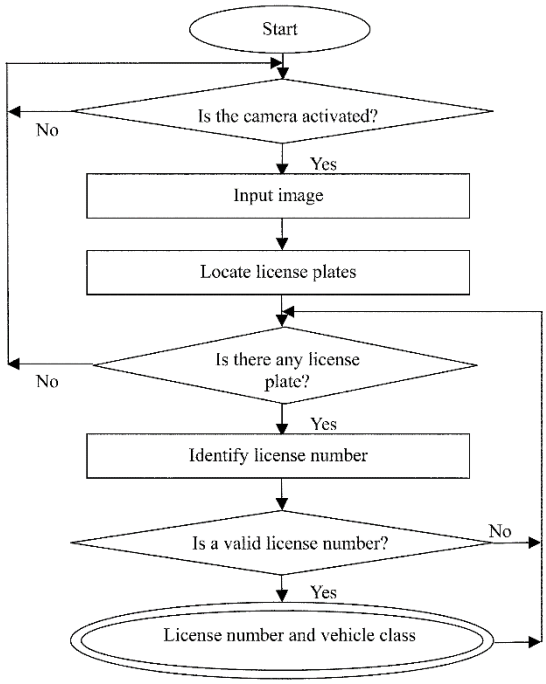

Figure 1 System Diagram

3.1 Dataset

While a large number of natural-image datasets exist for use in development of computer vision algorithms, few are devoted solely to the issue of license plate recognition.

We have chosen a license plate dataset created and maintained by the Department of Computer and Intelligent Systems at the University of Zagreb, Croatia. Unlike most such datasets which are used primarily for license plate localization and so contain poor quality images (since localization techniques do not require that the plate number itself be readable), this one contains much higher quality, close-up images that are far better suited to the task at hand. The pictures have been taken using an Olympus C-2040 digital camera at a resolution of 640x480, and the entire set contains over 500 images of various vehicles (cars, trucks, buses), taken under various lighting conditions (sunny, cloudy, rainy, twilight, night light).

A few examples are visualized in figure 2 below.

Figure 2 Sample Images

3.2 Preprocessing

Multiple levels of pre-processing are used to transform the images into viable subjects for image processing.

First, the image is converted into the grey-scale colour space since the task at hand does not require use of the extra colour channels. Furthermore, histogram equalization of the image is performed in order to boost contrast and improve overall picture quality for later steps.

3.3 License Plate Localization

To accurately determine the location and size of the car license plate, a variety of sanity checks are used in addition to basic filters.

We know that a car license plate is usually distinctly visible in an image, which means that strong changes in intensity must exist between the car body and the plate itself. Only then can it be easily identified by traffic wardens.

To take advantage of this artefact, we use gradient-based methods to outline the plate.

First, a 3x3 Gaussian filter is convolved over the image in order to get right of low level noise and detail. A Laplacian filter is then applied onto this blurred image, which highlights sharp changes in grey-levels across the image.

To locate the plate, we find all the contours that exist in the Laplacian image. This returns bounding boxes over all contiguous components of the image. We use certain thresholds on the area and aspect ratio of contours in order to narrow down the field. The width-to-height ratio of a license plate is roughly equivalent to 4.5. We accept any contours that match this figure up to a tolerance of 0.5.

Since, unlike most components in an image, a license plate is strictly rectangular in shape, it can be thought of as taking up a large percentage of the area covered by the box bounding it. This is known as the solidity and is calculated as a ratio of contour area to hull area. A number of trials performed over a range of solidity values produced 0.8 as the optimum value.

After all these steps, 90% accuracy in license plate detection and localization was achieved. The results of the Laplacian of Gaussian filter are shown below, followed by an example of accurate plate localization (represented by the bright green bounding box).

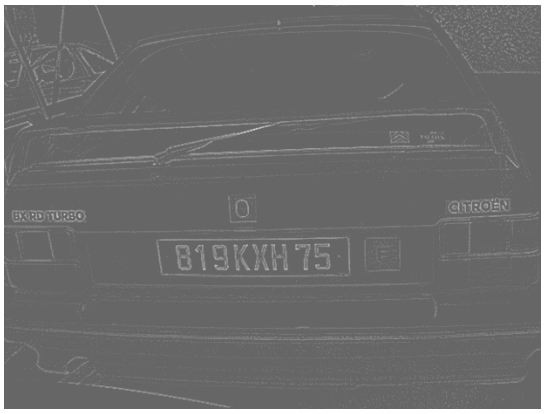

Figure 3 Laplacian of Gaussian

Figure 4 Localized License Plate

3.4 License Number Recognition

The task of recognizing the individual figures on the plate detected in the previous step proves rather difficult to solve solely via the use of image processing and morphological filters. Since license plates are alphanumeric, and the digits may occur in any order, it was deemed appropriate to make use of machine learning techniques to solve this problem.

Machine learning involves the application of algorithms that can improve their performance on a given task by repeated computation on some amount of data that is known to come from a similar probability distribution.

For this classification, we employ a K-Nearest Neighbour classifier, a non-parametric method used that can efficiently make use of a small amount of training data.

In k-NN classification, the output is a class membership. An object is classified by a majority vote of its neighbours, with the object being assigned to the class most common among its k nearest neighbours (k is a positive integer, typically small). If k = 1, then the object is simply assigned to the class of that single nearest neighbour.

To create a dataset for classification, we first segment alphanumeric characters from localized license plates and manually label them as one of 36 classes using a keyboard as input.

To segment the numbers from a plate, we use Otsu's Method to adaptively threshold the image using the mean of all its grey-scale intensities as the threshold value.

Figure 5 Thresholding for Segmentation

The process of contour discovery and selection is then repeated for these digits, this time using different thresholds for minimum area and aspect ratio. Each digit crop is resized to a 20x20 matrix and later flattened out into a 1x400 vector and appended to the dataset. The ASCII values entered via the keyboard are stored separately as labels.

A total of 380 alphanumeric characters are collected in this fashion. Since these are already randomly distributed, the first 300 are used for training the classifier while the remaining 80 are kept aside for cross-validation of the algorithm's performance.

A series of tests run with range of values for k (the number of neighbours to use for classification) resulted in k=1 being the optimum value. Using just a single neighbour gave an accuracy of 80%, whereas using 3 or 5 neighbours gave an accuracy of 77% and 74%, respectively.

4 Results

Once the system described above had been built, we performed end-to-end test to judge the performance of the entire pipeline.

A few results are shown below, with the detected plates outlined and the recognized number printed directly underneath.

Figure 6 Result 1

Figure 7 Result 2

Figure 8 Result 3

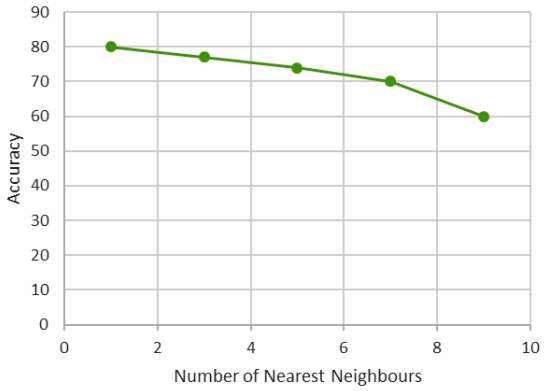

Figure 9 Effect of K on Accuracy

5 Discussion

As can be seen from the results displayed, our system achieves a high degree of accuracy on the task. However, it must be mentioned that there were quite a few cases in which either a) the plate localization step failed due to breaks in the gradient image, b) only a fraction of the characters on the plate were detected due to low illumination, or c) multiple characters were incorrectly detected at the same position on the page due to the existence of hierarchical contours.

6 Conclusion

In this paper we proposed, conceived, implemented, and evaluated a system for automatic license plate recognition. The development of this system involved a wide variety of image processing techniques, coupled with a machine learning algorithm, in order to provide maximum accuracy. While certainly not perfect, we find our results to be highly encouraging.

For future development, we hope to incorporate real-time surveillance into our system by passing frames of a video stream through all the steps discussed above and overlaying the car number back onto the video.

In addition to this, such a system may also be linked to a database of motorists with the additional functionality of querying the database and providing law enforcement officers with instant reports on the criminal status driver or the location of origin of the car.

These are only a few of the possible pathways of work that can be pursued in the future, and we are positive that the development of a holistic license plate recognition system can greatly benefit the government and concerned authorities of any city or town.

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